

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (CE/IT/EC)

May 2014

CL105 Mechanics of Structures

Date: 24.05.2014, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 Find centroid of the shaded area shown in Figure 1. [07]
- Q - 2 (a) The force F_1 is of 100N magnitude. The resultant of the force F_1 and unknown force F_2 is 100N in magnitude. Find the magnitude of F_2 if resultant is perpendicular to F_1 . [04]
- (b) Attempt any Two. [10]
- (i) A 50kN is the resultant of four forces out of which three are shown in Figure 2. Find the direction and magnitude of the fourth force.
- (ii) Two cylinders, each of diameter 40cm and mass 500N, rest in a rectangle channel of base width 72cm. Find the reactions at all contact surfaces. Refer Figure 3.
- (iii) Determine the resultant of the four forces shown in Figure 4 with respect to point B.
- Q - 3 (a) Convert 5×10^5 to kN/m^2 . [02]
- (b) Two tensile forces 80kN and 70kN acting at a point with angle 60° between them. Find magnitude and direction of the resultant. [02]
- (c) Find the moment of force 30kN about point 'O' shown in Figure 5. [03]
- (d) Determine the resultant of a co-planar concurrent force system if following forces are acting at a point: [07]
1. 100 kN tensile making an angle of 10° with +X axis.
 2. 200 kN tensile making an angle of 110° with +X axis.
 3. 300 kN tensile making an angle of 210° with +X axis.
 4. 400 kN tensile making an angle of 310° with +X axis.

SECTION – II

- Q - 4 Define stiffness, toughness and brittleness. Explain stress-strain curve of a ductile material when subjected to a tensile loading up to failure. [07]
- Q - 5 (a) Derive $\Delta l = 4PL/\pi E d_1 d_2$ with usual notations. [04]
- (b) Attempt any Two. [10]
- (i) A steel rail is 12m long and is laid at a temperature of 18°C . The maximum temperature expected is 40°C .
- a. Estimate the minimum gap between rails to be left so that the temperature stresses do not develop.
 - b. Calculate the temperature stresses developed in the rails if:

1. No expansion joint is provided.
2. If a 1.5 mm gap is provided for expansion

- (ii) Find out force P required for equilibrium of the bar shown in Figure 6. Find out total deformation of the bar if $E_s = 200\text{GPa}$, $E_b = 100\text{GPa}$, $E_{al} = 75\text{GPa}$.
- (iii) A bar of brass 20mm is enclosed in a steel tube of 40mm external diameter and 20mm internal diameter. The bar and the tubes are initially 1.2 m long and are rigidly fastened at both ends using 20mm diameter pins. If the temperature is raised by 60°C , find the stresses induced in the bar, tube and pins.

Q - 6 (a) A uniform ladder of length 4m placed against a vertical wall making an angle of 45° with the floor. The coefficient of friction between the wall and the ladder is 0.4 and that between floor and ladder is 0.5. If a man whose weight is $\frac{1}{2}$ of that of ladder ascends it, at what height will he be when the ladder slips? [07]

Q - 6 (b) Find out support reactions for **any one beam**. Refer Figure 7 & 8. [07]

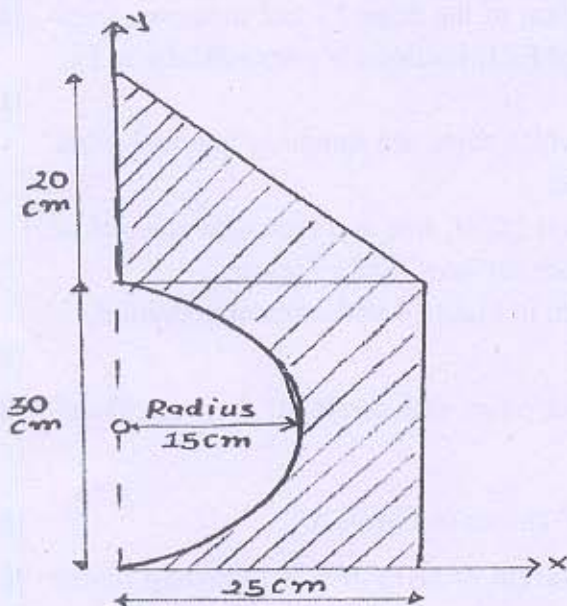


Figure 1

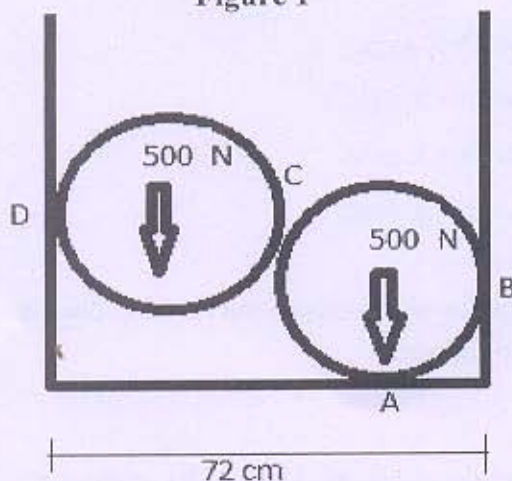


Figure 3

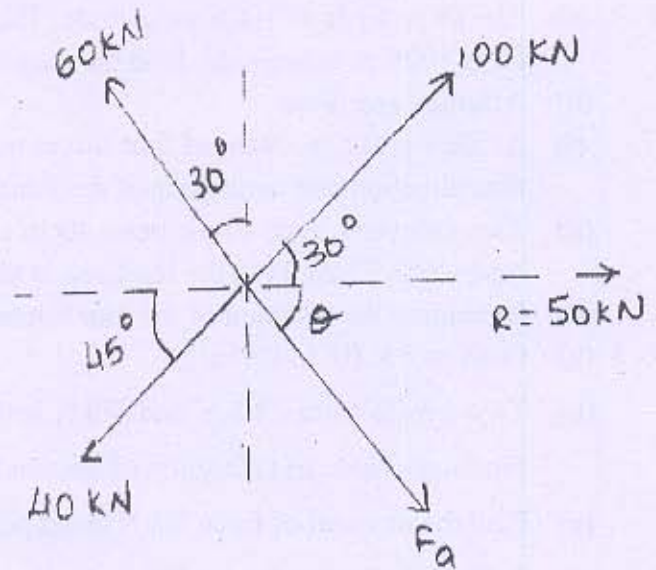


Figure 2

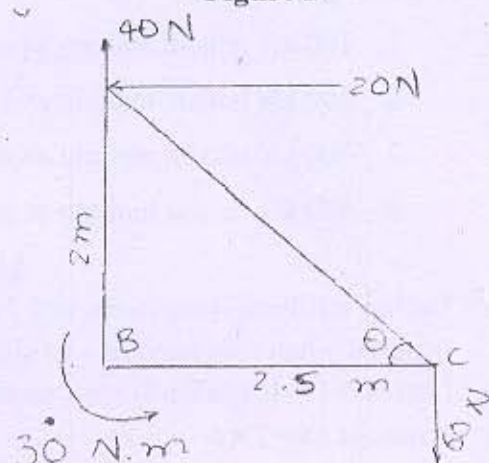


Figure 4

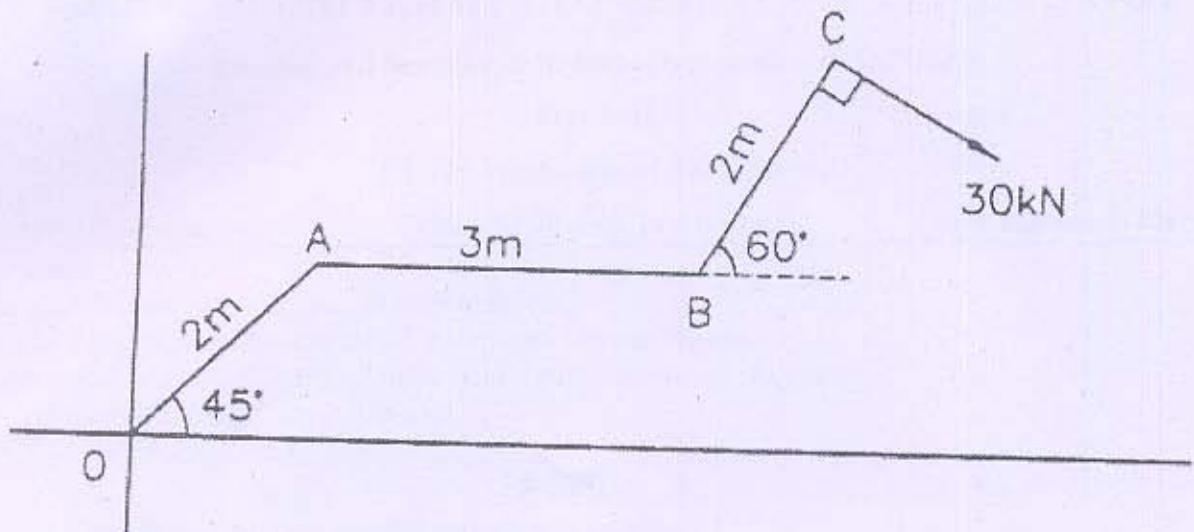


Figure 5

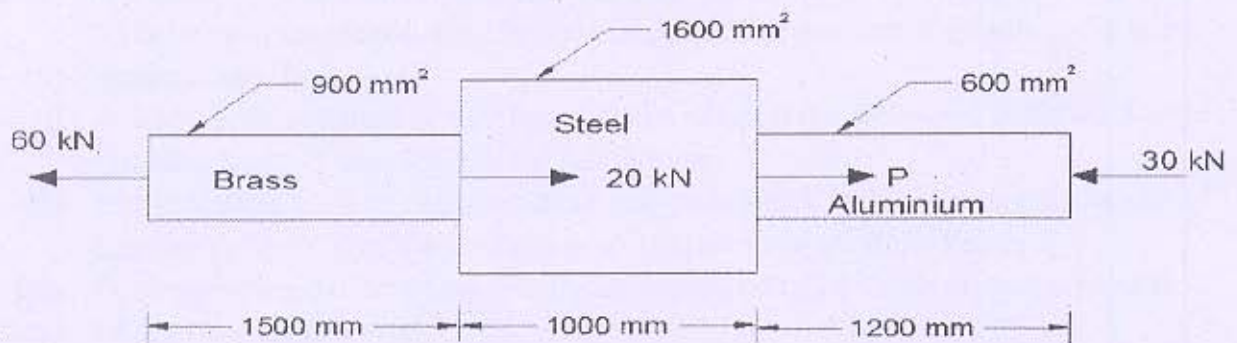


Figure 6

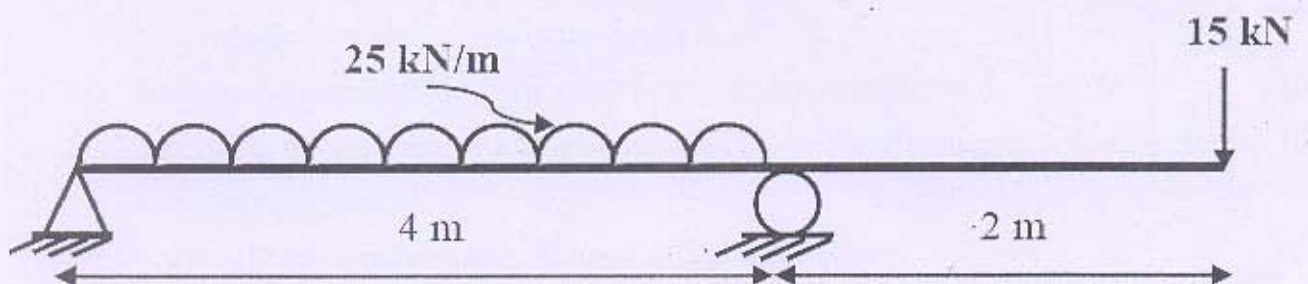


Figure 7

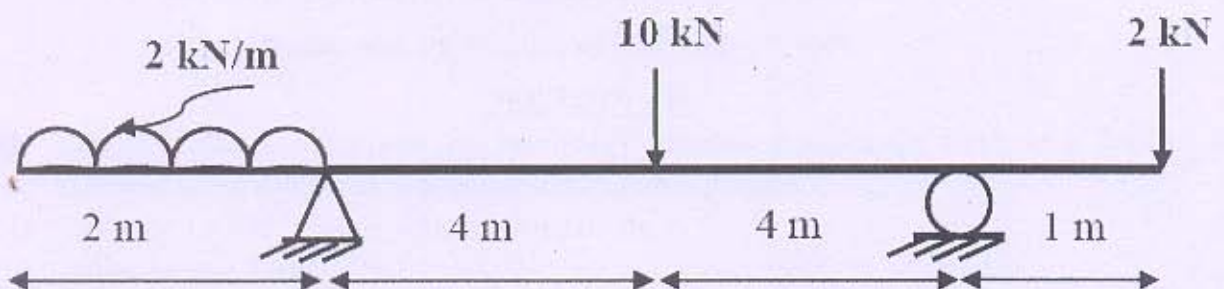


Figure 8
